

ICT and Society: Digital Media and Applications Part B. Digital Image, Graphics, and Multimedia

I. The Components of Multimedia is not simply the presence of media; it is the synchronized combination of text, audio, still images, animation, video, and interactivity. Its primary purpose in ICT is to enhance communication by engaging multiple senses simultaneously, making complex information easier to retain.

- **Text:** Despite the rise of visual media, text remains the fundamental anchor of multimedia. It provides the specific details, definitions, and narrative structure that images alone cannot convey. In web design and software, text serves a functional role through menus, buttons, and headlines, guiding the user through the experience.
- **Graphics (Images):** Graphics are used to break up large blocks of text and illustrate concepts that are difficult to describe verbally. Whether they are photographs, charts, or user interface icons, graphics process in the human brain much faster than text, providing immediate context or emotional background.
- **Audio:** Sound adds a layer of emotional depth and realism. It functions in three ways: **background audio** sets the mood, **sound effects** provide feedback (like a "ding" when a task is finished), and **voice-overs/narration** provide instructional guidance, often used in eLearning to support auditory learners.
- **Video:** Video is the technology of capturing and reconstructing moving pictures. It is the most bandwidth-intensive component but also the most powerful for storytelling. It combines visual and auditory elements to provide a realistic experience, making it ideal for tutorials, news, and entertainment.
- **Animation:** Unlike video, which captures real life, animation is the illusion of motion created by rapidly displaying a sequence of static, artificial images. It is particularly useful for explaining abstract concepts or internal processes—such as showing how a car engine works internally—where a physical camera cannot go.

II. Digital Image and Graphics Computers store images in two fundamentally different ways. Understanding the difference between these two types—Raster and Vector—is critical for anyone working in design or ICT, as it dictates how an image can be used.

- **Raster Graphics (Bitmap):** Raster images are composed of a fixed grid of tiny colored dots called **pixels**. When you look at a photograph on a computer, you are looking at a raster image. The defining characteristic of raster graphics is that they are **resolution-dependent**. Because there is a fixed number of pixels, zooming in too far causes the image to look blocky or "pixelated."

Context on Formats: We use **JPEG** for photos because it compresses complex colors well. We use **PNG** when we need transparent backgrounds (like a logo floating over a video). We use **GIF** for simple, looping animations.

- **Vector Graphics:**

Unlike raster images, vector graphics do not use pixels. Instead, they store instructions based on **mathematical formulas** (e.g., "draw a circle with a radius of 5 at these coordinates"). Because they are math-based, they are **resolution-independent**. You can scale a vector image from the size of a business card to the size of a billboard, and it will remain perfectly crisp. This makes vectors the standard for logos, typography, and technical blueprints (**SVG, AI, EPS**).

- **Color Models (Context of Usage):**

RGB (Red, Green, Blue): This is an **additive** color model used for anything that involves light (monitors, phone screens, projectors). It combines colored light to create white.

CMYK (Cyan, Magenta, Yellow, Key/Black): This is a **subtractive** color model used for physical printing (ink). It combines inks to create dark colors. A common mistake in ICT is designing in RGB and printing in CMYK, which results in dull, incorrect colors.

III. Digital Audio and Video To process sound and motion, computers must convert continuous analog waves (like sound in the air or light hitting a lens) into discrete digital binary code.

- **Digital Audio:** Digital audio works by taking "snapshots" of sound waves thousands of times per second. This is called the **Sampling Rate**. A standard CD quality is 44.1 kHz, meaning the audio was sampled 44,100 times per second. Higher sampling rates capture more detail but result in much larger file sizes.

Compression Context: Because raw audio files are huge, we compress them.

Lossy compression (like **MP3**) deletes sounds the human ear can barely hear to save space. **Lossless compression** (like **FLAC**) shrinks the file mathematically without throwing away any data, preserving perfect quality.

- **Digital Video:** Video is essentially a slideshow played at high speed. The **Frame Rate (FPS)** dictates how smooth the motion looks. Standard movies use 24fps for a cinematic "blur," while video games and sports broadcast at 60fps+ for crisp, fluid motion.

Codecs vs. Containers: This is often confused. A **Container** (like **.MP4** or **.AVI**) is just the box that holds the files. The **Codec** (like **H.264** or **HEVC**) is the software inside the box that squeezes (compresses) the video data to make it fit. You need the right codec installed to play the video inside the container.

IV. Graphics and Multimedia Applications Different software tools are specialized for different stages of the multimedia workflow.

- **Image Editing Software:**

Raster Editors (Photoshop/GIMP): These tools work by altering pixels. They are like a digital darkroom. You use them to retouch photos, color-correct lighting, or blend images together. They are essential for photographers and advertisers.

Vector Editors (Illustrator/CorelDRAW): These tools work by manipulating lines and shapes. They are like a digital drafting table. You use them to create logos, icons, and layout designs that need to be scalable for branding.

- **Audio/Video Editing Software:**

Non-Linear Editors (NLE): Software like **Adobe Premiere** or **DaVinci Resolve** allows editors to arrange video clips on a timeline non-sequentially (hence "non-linear"). They handle the visual storytelling, color grading, and transition effects used in film and YouTube content.

Digital Audio Workstations (DAW): Tools like **Audacity** or **Pro Tools** are used strictly for audio. They function like a recording studio console, allowing engineers to mix multiple tracks (vocals, drums, bass) into a single song or podcast.

- **Multimedia Authoring Tools:** Once all the assets (images, video, audio) are created, authoring tools like **Unity** or **Adobe Animate** are used to assemble them into a cohesive, interactive experience. These tools add the "logic" or programming that allows a user to click a button and trigger a sound or move a character in a game.